

LEDGER · 2026-09-04 · AGRONOMY / MATERIALS SCIENCE

Paraffin-Functionalized Sand (PFS) Mulch via Thermal Rotary Encapsulation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Polyethylene (PE) Agricultural Film
Entry application	High-Value Arid Horticulture (Sweet Peppers and Tomatoes)
Measured advantage	Total Fruit Yield (Sweet Pepper): +112% yield vs bare soil for PFS Mulch compared to +71% for PE Film (1.23x over incumbent)
Time to first revenue	3 months
Gross margin at parity	76.0%
Startup CapEx	\$101,500
Payback from first sale	3.2 months
Capital productivity	7.09x

Regulatory risk

Low — no material regulatory CapEx reported (\$0 estimated)

Market absence

Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named

Why it failed

- Voided by: Negative control / sub-2x

Assessment

The concept demonstrates robust peer-reviewed science, sub-\$250k CapEx (\$101,500), fast time to revenue (3 months), and high gross margins (76%). However, it fails the strict superiority threshold because its primary quantified performance metric (fruit yield) shows only a 1.23x multiplier over PE film (+112% vs +71%), missing the mandatory 2x step-change bar. Environmental benefits like zero plastic waste and full biodegradability are preference attributes and cannot substitute for primary performance superiority.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

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